SSH Configuration and Remote Access

1. Aim

To configure **SSH (Secure Shell)** on a Cisco router in Cisco Packet Tracer and remotely access the router from a PC.

2. Introduction

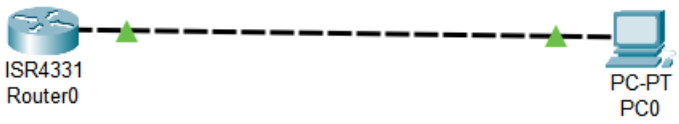
SSH is a network protocol used to securely access and manage a remote device through a command-line interface. It encrypts the communication between the client and server. SSH normally uses **TCP port 22**.

In this practical, **PC0 acts as the SSH client** and **Router0 acts as the SSH server**.

3. Network Topology

The network contains:

- **1 Cisco ISR4331 Router (Router0)**
- **1 PC (PC0)**
- A network connection between the PC and router



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4. IP Configuration

- The router and PC were configured in the same network.

Device	IP Address	Subnet Mask	Default Gateway
Router0	192.168.1.1	255.255.255.0	192.168.1.2
PC0	192.168.1.2	255.255.255.0	-

5. Router Configuration

First, the router interface was configured with an IP address:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet 0/0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
```

Then SSH was configured on the router:
Router(config)# hostname Router0
Router0(config)# ip domain-name example.com
Router0(config)# username admin secret cisco123
Router0(config)# crypto key generate rsa

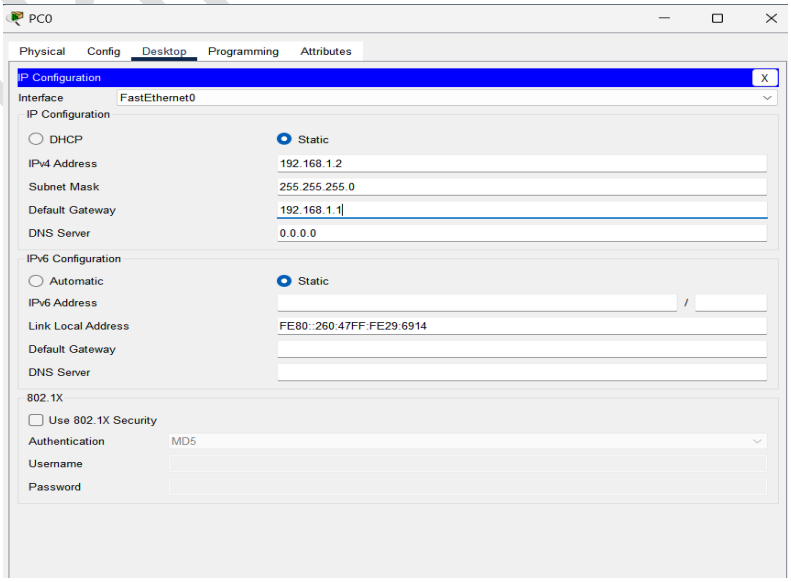
RSA keys are required to establish the secure SSH connection.
The VTY lines were then configured to allow SSH:

Router0(config)# line vty 0 4
Router0(config-line)# login local
Router0(config-line)# transport input ssh

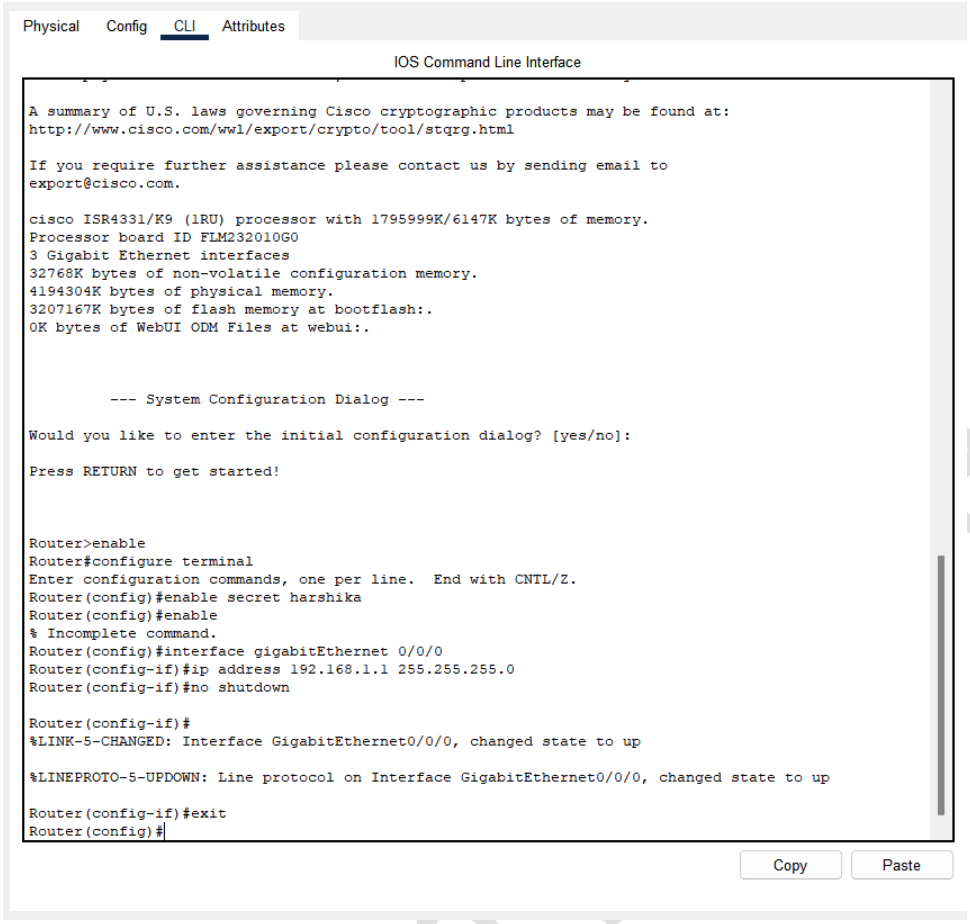
6. PC Configuration
PC0 was configured with:
IP Address: 192.168.1.2
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

7. Testing the Connection
The connection was first tested using the ping command:
PC> ping 192.168.1.1
A successful ping confirms that PC0 can communicate with Router0.
The SSH connection was then started using:
PC> ssh -l admin 192.168.1.1
After entering the correct password, the PC was connected to Router0 through SSH.

Figure 2: Successful SSH login



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```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#enable secret harshika
Router(config)#enable
% Incomplete command.
Router(config)#interface gigabitEthernet 0/0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

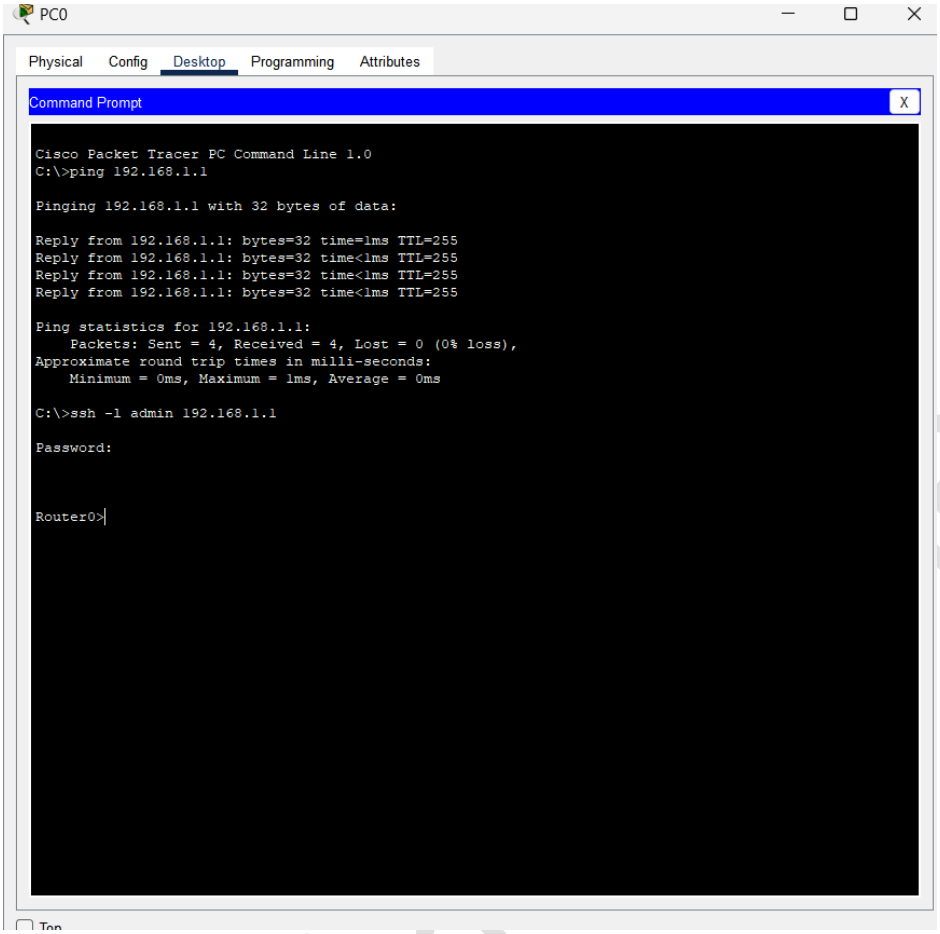
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

Router(config-if)#exit
Router(config)#hostname Router0
Router0(config)#ip domain name example.com
Router0(config)#username admin secret cisco123
Router0(config)#crypto key generate rsa
The name for the keys will be: Router0.example.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

Router0(config)#line vty 0 4
*Mar 1 0:8:27.273: %SSH-5-ENABLED: SSH 1.99 has been enabled
Router0(config-line)#login local
Router0(config-line)#transport input ssh
Router0(config-line)#
```



8. Result

The SSH connection between **PC0 and Router0** was successfully established. PC0 was able to remotely access Router0 using SSH. This demonstrates how SSH can be used for secure remote management of a network device.

9. Conclusion

In this practical, SSH was configured on a Cisco ISR4331 router using Cisco Packet Tracer. PC0 was configured as the SSH client and Router0 as the SSH server. The connection was tested using ping and SSH login. The practical demonstrated how a network administrator can securely access and manage a router remotely.

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